Tracking Volatility in Federal Grant Opportunities

Emily Stern
EPPS 6354 Information Management - Spring 2026

The Problem

- Since the 2025 administration change, grant funding revisions, freezes, and removals have accelerated at unprecedented rates (Clegg 2025)
- Nonprofits and local governments serving vulnerable populations have experienced measurable disruptions to operations and service delivery (Urban Institute 2025)
- Data governance scholars argue auditable public data is a matter of democratic accountability (OECD 2022; Osakwe 2025)

Why a Database is Needed

- No public historical record exists of federal grant changes over time
- Grants.gov only shows current-state snapshots — removed opportunities disappear without a trace
- A relational database can capture what Grants.gov doesn't: when opportunities appeared, changed, and disappeared

System Design

- Automated web scraping captures periodic grants.gov snapshots
- Snapshot-to-snapshot comparisons detects additions, modifications, and removals in opportunities, linking with primary key.
- Database stores a copy of the current grants.gov instance, along with a full history of modified opportunities in the change log.
- Front End: R Shiny app for querying opportunities and visualizing trends

Entity Relationship Diagram

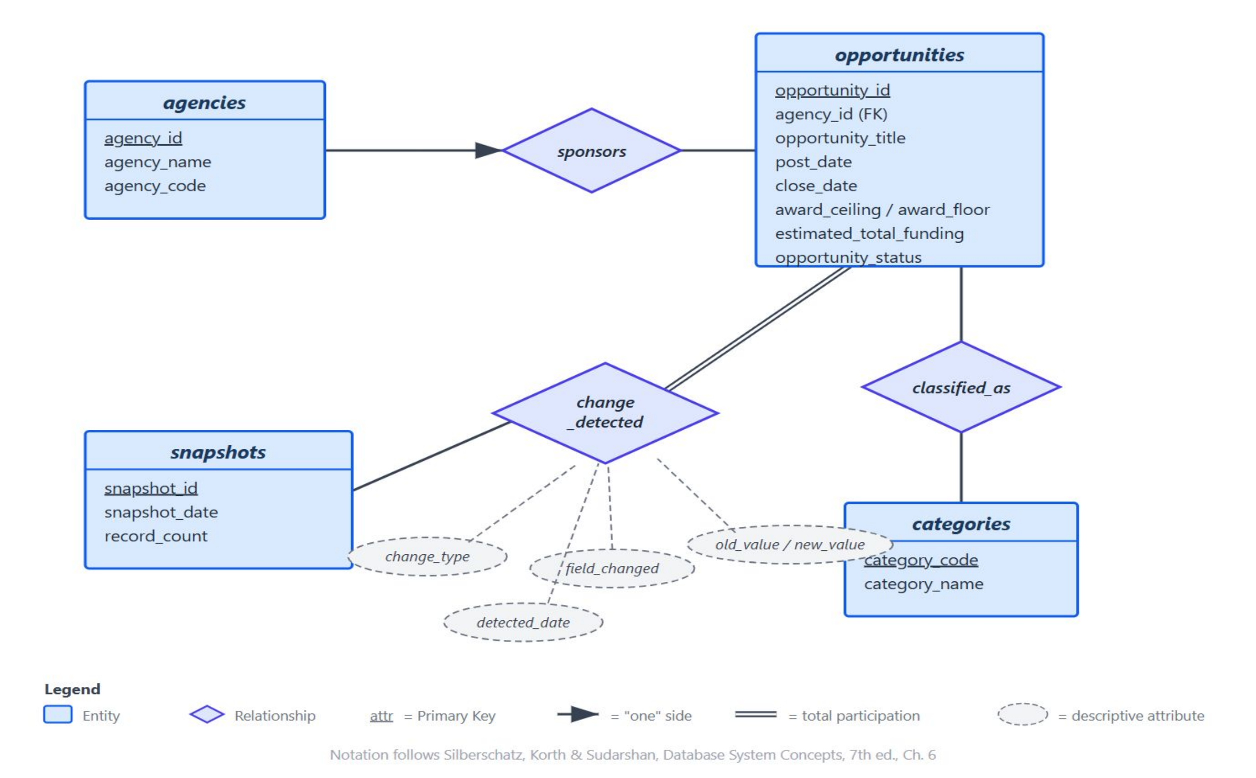
Entities (rectangles):

agencies — reference table for federal agencies; surrogate primary key (agency_id)

opportunities — core table storing each grant's data; foreign key links to its sponsoring agency

categories — funding category lookup using Grants.gov's own type codes as a natural primary key

snapshots — audit metadata recording when each Grants.gov extraction occurred and how many records it contained



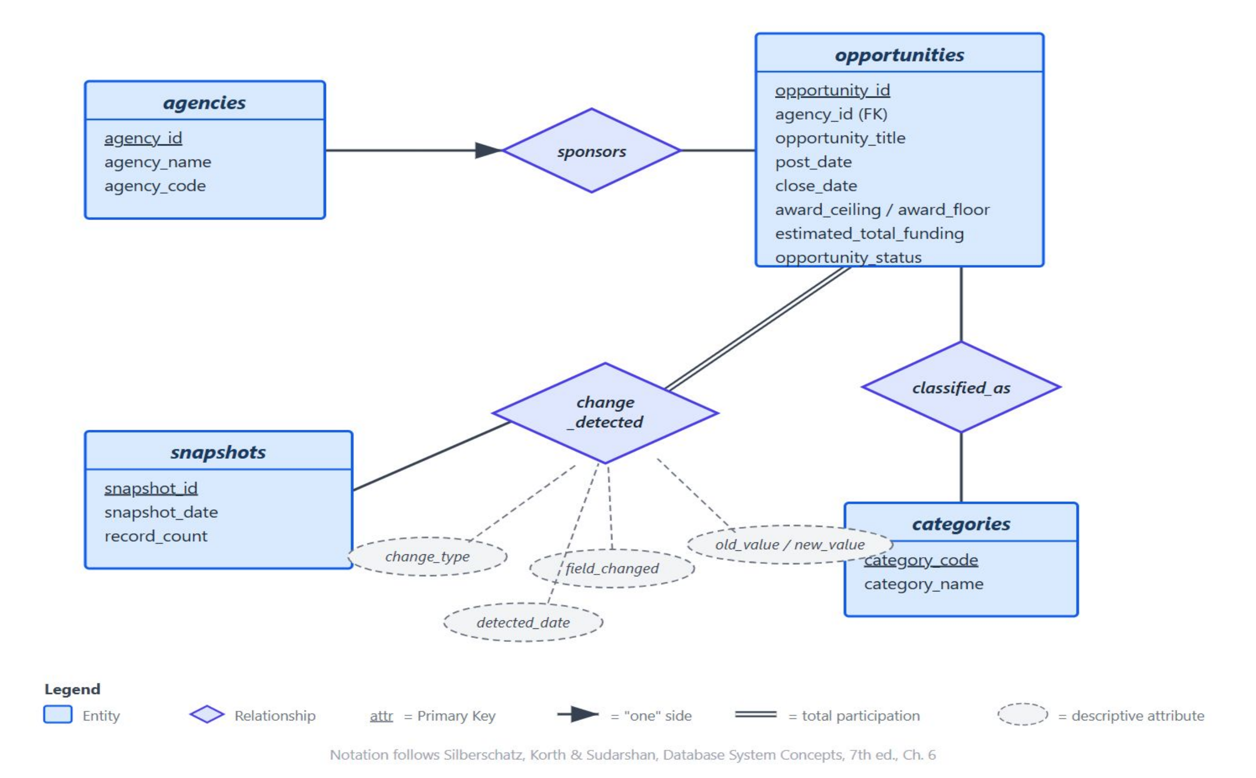
Entity Relationship Diagram

Relationships (diamonds):

sponsors — one agency can sponsor many opportunities; each opportunity belongs to exactly one agency

classified_as — many-to-many relationship; a grant can fall under multiple funding categories and a category contains many grants

change_detected — links opportunities to snapshots, recording what changed between extractions.



Tech Stack & Next Steps

- PostgreSQL, R (rvest/httr2 for scraping, Shiny for app, DBI/RPostgres for connectivity), GitHub for hosting
- Next steps: finalize scraping pipeline, populate initial snapshots, build out Shiny dashboard, validate change detection logic

AI Use Disclosure

This project made use of generative artificial intelligence tools, specifically OpenAI's GPT models and Anthropic's Claude models, to support the writing and refinement process. These tools were used to assist with literature review, project planning, structural organization, and stylistic editing of draft text. All substantive intellectual contributions—including the selection of the research topic, formulation of the research motivation, database schema design, methodological approach, interpretation of policy context, verification of sources, and final wording—were produced and evaluated by the author.

References

Anthropic. (2024). Claude [Large language model]. https://www.anthropic.com

Clegg, K. A. (2025). Research implications of increased political oversight in the United States. Journal of Medical Internet Research, 27, e83499. https://doi.org/10.2196/83499

OpenAI. (2024). GPT-4 technical report. https://openai.com/research/gpt-4

Organisation for Economic Co-operation and Development. (2022). Going digital: A guide to data governance for policy making. OECD Publishing. https://doi.org/10.1787/40d53904-en

Osakwe, J. (2025). Data governance in the public sector: Enhancing accountability and service delivery. Public Administration Review, 11, 1–11.

Urban Institute. (2025, October). How government funding disruptions affected nonprofits in early 2025. https://www.urban.org/sites/default/files/2025-10/How_Government_Funding_Disruptions_Affected_Nonprofits_in_Early_2025.pdf